QuantiFi 2026 — Strategy Abstract

Team 25

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1 Introduction

We trade all 26 products: six core products, `LYCHEE` with its ten call vouchers and four fruit products, and five pearls. Each strategy is built on a pattern we measured in the data; where we found none, we quote small with a tight position cap. One `Trader` class passes each tick to three strategy classes (one per part below), gives each its own memory, and clips every order to the position limit, so one strategy can never crash or overwrite another. To avoid overfitting, we checked every strategy on held-out days (tune on two days, score the third), and the main ones also on shifted, mirrored and pattern-free test days (Section 5). The final bot makes **409,590** over days −2 to 0 (Figure 1).

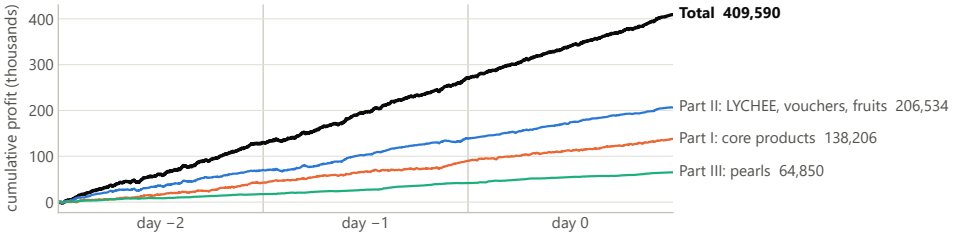
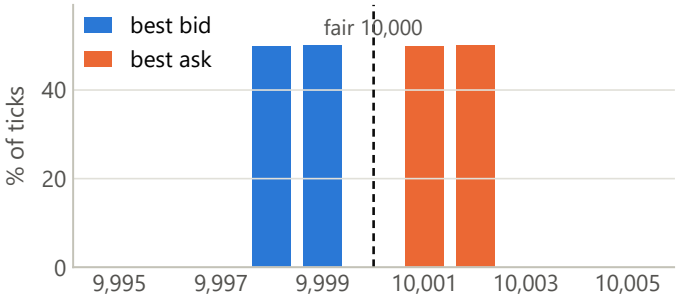
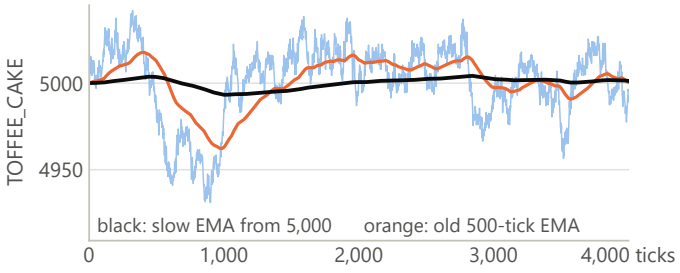


Figure 1: Cumulative profit of the final bot over the three training days, by part.

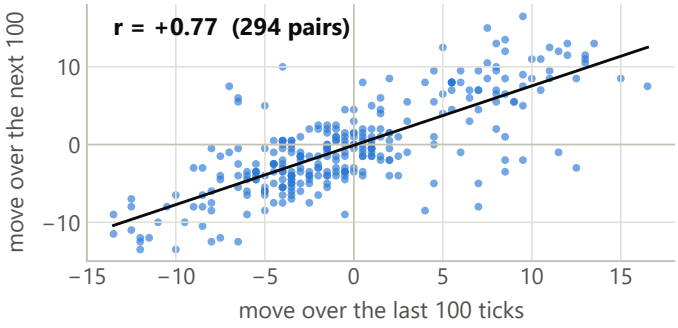
2 Part I: Core Products



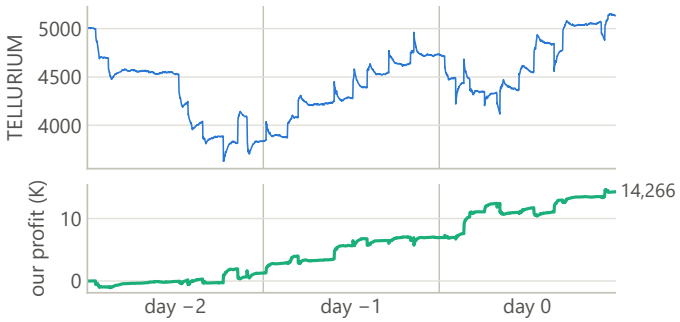
(a) `AMETHYST`: best bid and ask never cross 10,000.



(b) `TOFFEE_CAKE`: slow anchored EMA vs. the old fast one.



(c) `PAPER_LANTERNS`: last vs. next 100-tick move.



(d) `TELLURIUM`: price jumps (top), our profit (bottom).

Figure 2: The patterns behind Part I, from the three training days.

Each tick we estimate a fair value $\hat{p}_t$ (usually an EMA of the mid price), **take** any quote at least θ through it, and **make** with what is left: either *join* the best bid and ask, or post a *skewed quote* around $\hat{p}_t - \kappa q_t$, which leans away from our position q_t.

Product	Pattern	Fair value $\hat{p}_t$	Execution	Cap	Profit
AMETHYST	fixed at 10,000	10,000	take + join	50	1,613
TOFFEE_CAKE	snaps back to 5,000	EMA N=4000 from 5,000	take + skewed	50	65,395
PAPER_LANTERNS	trends persist	Eq. (1)	take + join	50	55,466
TELLURIUM	random, big jumps	none	join only	6	14,266
JELLY, PEANUT_BUTTER	random walks	EMA N=50 / 20	skewed	3	1,466

Table 1: Part I at a glance. A cap far below the exchange limit means there is no edge, so we hold little.

TOFFEE_CAKE: mean reversion. Its daily average was 5,002–5,004 every day. A 500-tick EMA chased the price and missed half of each reversion; a very slow EMA that starts at 5,000 stays near home but would still follow a new level (Figure 2b).

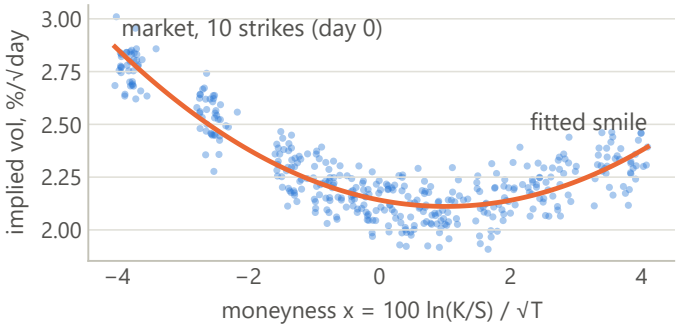
PAPER_LANTERNS: momentum. A 100-tick move predicts the next one ($r = +0.77$, Figure 2c), so we price the trend into fair value, with e_t a 20-tick EMA:

$$\hat{p}_t = e_t + 0.75\,(e_t - e_{t-100}). \tag{1}$$

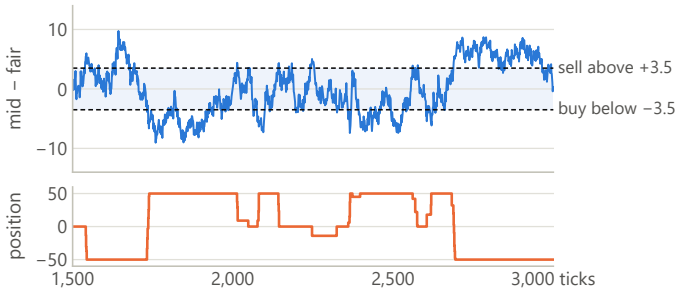
This doubled profit over an EMA crossover (27.9K → 55.5K).

AMETHYST, TELLURIUM, JELLY, PEANUT_BUTTER. AMETHYST's fair value is a fixed 10,000, so we take anything through it and join the rest. The other three have no pattern and TELLURIUM jumps, so we only earn the spread with small caps; its profit still climbs steadily (Figure 2d).

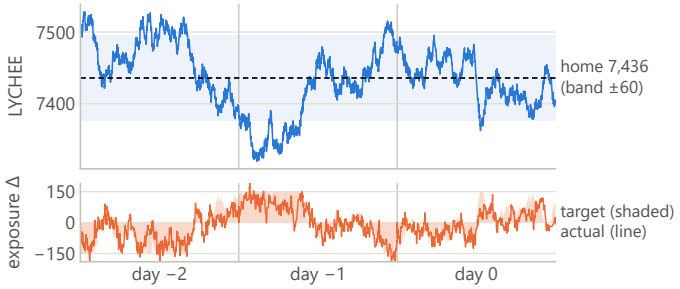
3 Part II: LYCHEE Options & Fruits



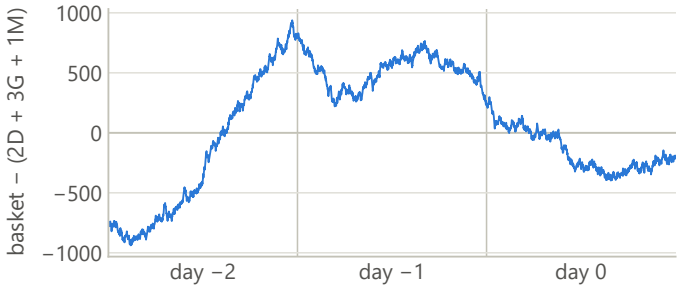
(a) Implied vols of all ten strikes on one smile.



(b) VOUCHER_7400: price − fair, and our position.



(c) LYCHEE around home, and our exposure Δ.



(d) Basket minus its fruits: the gap drifts.

Figure 3: The patterns behind Part II. Panels (b) and (c) use the bot's own fair values and positions.

Vouchers: Black–Scholes and the smile. The ten vouchers are calls on LYCHEE (price S), strikes K = 6,800–8,100, limit 50 each. Every tick we invert Black–Scholes

$$C(S,K,T,\sigma) = S\,\Phi(d_1) - K\,\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S/K) \pm \tfrac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \tag{2}$$

to get each voucher's implied volatility, then fit one smile through all ten strikes and smooth it over time (200-tick half-life):

$$\sigma(x) = a + b\,x + c\,x^2, \qquad x = 100\ln(K/S)/\sqrt{T}. \tag{3}$$

A single voucher misses the smile by only 0.09 vol points, and the miss fades (Figure 3a). Fair value is $\hat{C}_K = C(S, K, T, \sigma(x_K))$. We buy at $\hat{C}_K - 3.5$ and sell at $\hat{C}_K + 3.5$, then close the position once the price has moved 1.5 past fair the other way (Figure 3b). **Vouchers: 197,062.**

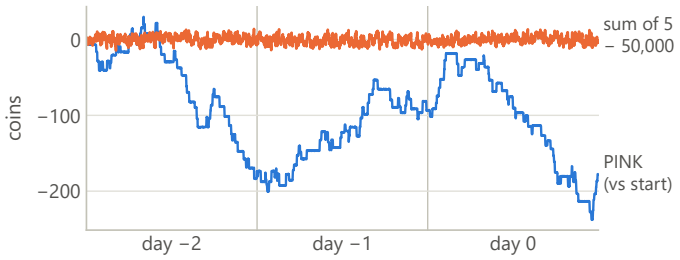
The 150 lean. Our total `LYCHEE` exposure is $\Delta = q_{\text{LYCHEE}} + \sum_K q_K \, \Phi(d_1^K)$. `LYCHEE` drifted back toward $H \approx 7{,}436$ every day, so we shade the voucher fair values to steer Δ toward

$$\Delta^* = -150 \cdot \text{clip}\left(\tfrac{S-H}{60}, -1, 1\right) \tag{4}$$

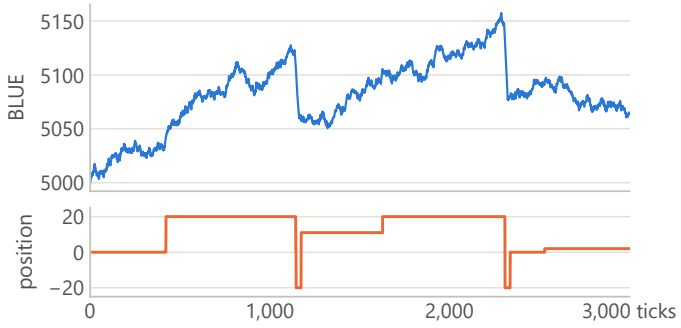
(Figure 3c). This is a bet: on held-out days it added 10–14K a day, but a day on which `LYCHEE` drifts away would cost 13–15K, so we use 150, not the 300 the backtest liked. The lean adds about 32K over the three days, mostly through the vouchers; `LYCHEE` itself makes **9,018**.

Fruits and basket. The basket is close to 2 `DRAGONFRUIT` + 3 `GRANADILLA` + 1 `MANGOSTEEN`, but the gap drifts by hundreds (Figure 3d), so there is no safe arbitrage. We join the best bid and ask with a cap of 1.

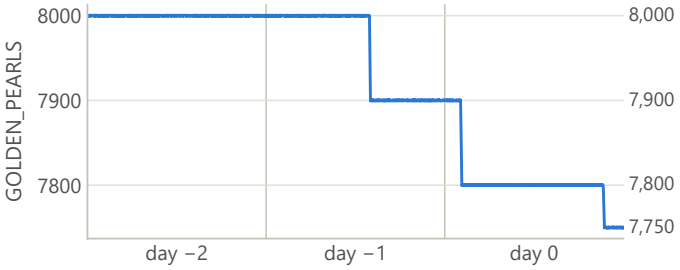
4 Part III: Pearls



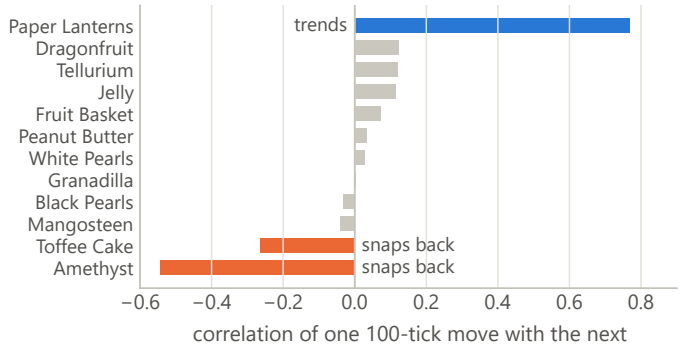
(a) `PINK` wanders; the sum of all five does not.



(b) `BLUE` sawtooth and our position.



(c) `GOLDEN`: flat levels, short slides.



(d) Only 3 of 12 products remember their last move.

Figure 4: The patterns behind Part III. Panel (d) is why every random walk gets small quotes.

PINK: the sum constraint. The five pearl mids always sum to 50,000 (sd 5), and `PINK` is the one that closes any gap: 70% of it within 50 ticks (Figure 4a). So

$$\text{fair}_{\text{PINK}} = m_{\text{PINK}} + 0.5\Big[\big(S^* - \textstyle\sum_{j \neq \text{PINK}} m_j\big) - m_{\text{PINK}}\Big], \tag{5}$$

where S^* starts at 50,000 and slowly follows the real sum. **27,098.**

BLUE: sawtooth. `BLUE` creeps up, then drops ∼45 in under 50 ticks (29 times in 3 days). We ride any move of 12+ within 10 ticks at the full limit (±20) until it stalls, and hold the creep in between (Figure 4b). **27,747.**

GOLDEN: step levels. Flat at 8,000, then 7,900, 7,800 and 7,750, with short slides between (Figure 4c). We quote 1 around the level (cap 5), and ride a slide once the price leaves the level by more than 3. **4,886.**

WHITE and BLACK. Random walks (Figure 4d), so small quotes with caps of 10 and 1. **5,120.**

5 Results and Validation

	Day −2	Day −1	Day 0	Total
Part I: Core	42,396	47,717	48,092	138,206
Part II: LYCHEE	70,519	68,680	67,334	206,534
Part III: Pearls	17,511	24,221	23,118	64,850
Total	**130,426**	**140,618**	**138,546**	**409,590**
Largest drop	6,255	6,870	3,399	
Sharpe-style ratio	5.8	6.4	6.6	

Table 2: Profit and risk on the training days.

- **Holds up on held-out days:** gains on hidden days match gains on tuning days (Figure 5). With a pattern removed, each pearl strategy loses at most ∼4K a day.

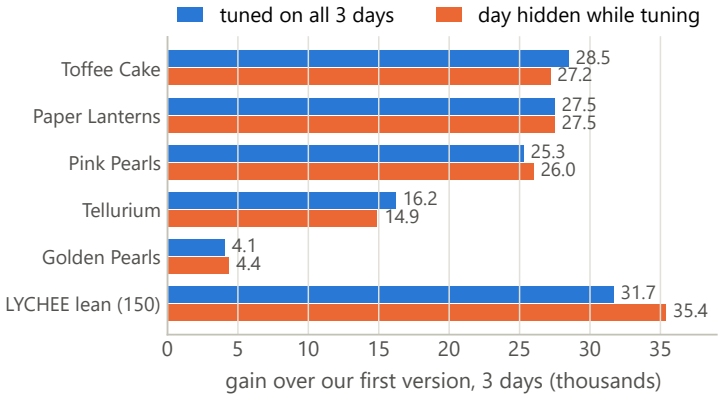


Figure 5: Rotation test: settings picked on two days and scored on the third, three times. Gains on the hidden day match gains on the tuning days.